

Phase3_Daily_Schedule

Phase 3 — Day-by-Day Schedule · Modern DL Theory + Foundation Architectures

Weeks 31-36 · Mon Apr 5 → Sun May 16, 2027 · ~165 hrs · → Tier 1 (broad)

Goal: stop treating transformers and diffusion as black boxes — derive every layer of attention, understand optimization dynamics, and defend architecture choices on theory. Feeds Phase 5 and the generative parts of the molecular track (6C).

Blocks: A 06:00–08:00 (theory) · B 08:30–10:30 (build + threads) · Evening 20:30–22:30 (papers, Anki, R). Weekend = buffer + rest. Threads: T DSA+C++ (Block B ~45 min) · M implement-from-scratch (Block B) · R research (Evening ~2 h/wk).

Week 31 · Apr 5-11 — Optimization dynamics

Day	Block A	Block B (T/M + build)	Evening (+ R)
Mon	Ruder: SGD → momentum	M: implement SGD + momentum from scratch	Anki
Tue	Adam family (RMSProp, Adam, AdamW)	M: implement Adam; compare on a 2D loss	Anki
Wed	Boyd: convex sets & functions	T: DP → visualize convex vs non-convex	Boyd lecture; Anki
Thu	Gradient/Newton; conditioning	M: LR-warmup + schedule experiment	Anki
Fri	Why DL trains despite non-convexity	T: timed set	Whiteboard-Fri: why momentum accelerates; what Adam's 2nd moment does
Sat-Sun	Buffer + rest		

Week 32 · Apr 12-18 — Information theory

Day	Block A	Block B	Evening
Mon	MacKay ch 1-2: entropy, information	T: graphs → implement entropy/CE/KL cleanly	Anki
Tue	KL divergence; cross-entropy from KL	M: KL between two Gaussians	Anki
Wed	Mutual information	T: DP → MI estimate on a toy set	Anki
Thu	MI in contrastive losses (InfoNCE)	M: a tiny InfoNCE loss	R: reproduce-paper
Fri	Coding & compression view	T: timed set	Whiteboard-Fri: derive cross-entropy from KL; NLL = MLE
Sat-Sun	Buffer + rest		

Week 33 · Apr 19-25 — Transformers in depth

Day	Block A	Block B	Evening
Mon	CS25: attention & the transformer	M: multi-head attention as one matmul	Annotated Transformer
Tue	Positional encodings; RoPE	M: sinusoidal + RoPE from scratch	Anki
Wed	RMSNorm; pre-LN vs post-LN	T: intervals → add RMSNorm to your block	Lilian Weng blog
Thu	FlashAttention; GQA/MQA (concepts)	M: SwiGLU FFN; assemble a full block	Anki; R: write-up
Fri	MoE / Switch (concept)	T: timed set	Whiteboard-Fri: derive attention; param-count a block
Sat-Sun	Buffer + rest · Deliverable: a from-scratch transformer block (attention+RoPE+RMSNorm+SwiGLU)		

Week 34 · Apr 26-May 2 — Vision transformers + modern ConvNets

Day	Block A	Block B	Evening
Mon	ViT lectures (patch embedding)	M: ViT patch-embed + classifier head	Anki
Tue	ConvNeXt (a modernized CNN)	T: graphs → run a ConvNeXt baseline	Anki
Wed	DINOv2 (SSL)	M: DINOv2 features + linear probe	Anki

Thu	MAE (masked autoencoding)	T: DP → an MAE-style masking demo	R: reproduce-paper
Fri	CNN vs ViT inductive biases	T: timed set	Whiteboard-Fri: why SSL helps in low-data (ties to imaging FMs)
Sat-Sun	Buffer + rest		

Weeks 35-36 · May 3-16 — Diffusion & generative modeling (feeds 6C)			
Day	Block A	Block B	Evening (+ R)
Mon (Wk 35)	CS236: latent-variable models, VAEs	M: a small VAE	Anki
Tue	Normalizing flows (concept)	T: DP → flow toy	Anki
Wed	DDPM: forward/reverse process	M: DDPM forward (noising)	DDPM paper
Thu	DDPM: the ϵ -prediction loss	M: DDPM reverse + ϵ -loss	Anki
Fri	Score-based view (Yang Song)	T: timed set	Whiteboard-Fri: derive the diffusion loss (both views)
Mon-Thu (Wk 36)	Guidance; DDIM sampling; conditioning	M: train the DDPM on a toy set; sampling; T: 1 set/day	HF Diffusion course; R: write-up
Fri (Wk 36)	🔗 PHASE-3 SELF-TEST (attention on whiteboard in 5 min; derive diffusion loss; LayerNorm-not-BatchNorm in transformers; positional encodings compared; param-count a block)	Polish a minimal-DDPM repo	R: post write-up
Sat-Sun	Buffer + rest · Deliverable: minimal DDPM (from scratch) trained on a toy set		

End of Phase 3 → Phase 5 (Production AI Systems + MLOps) next.